Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	*Cognitive Energy Autonomy: Neuro-Symbolic Agents for Physics-Informed Diagnostics and Active Flexibility Control*			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI (√), Next-gen Communication ( ), Advanced Robotics & Manufacturing (√), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Energy & Resources		*Mechanical & Materials*	
	Medium Category	Small Category	Medium Category	Small Category
	Power IT	*Intelligent Power Grid Platform Technologies*	*Clean Production*	*Environmentally Friendly Product Manufatcuring Technologies*
Concept of Proposed Technology	*We propose a **"Cognitive Energy Agent System"** that moves beyond passive monitoring to autonomous decision-making and active control in smart factories. The system is built on three pillars:* *1. **Physics-Informed Anomaly Detection (PINNs):** Integrating fluid dynamics laws into deep learning models to detect anomalies (e.g., compressed air leakages) and quantify physical losses with minimal sensor data.* *2. **Active Flexibility Control Loop:** A closed-loop system that combines probabilistic forecasting with mathematical optimization (MILP/Deep Reinforcement Learning). It actively steers production loads and storage systems to act as a **Virtual Power Plant**, reacting to volatile market prices in real-*			

	time. *3. **LLM-driven Decision Agent:** Synergizing **Time Series Foundation Models (TSFMs)** with **Large Language Models (LLMs)**. The LLM acts as the reasoning engine that interprets the outputs of TSFMs, orchestrates optimization workflows, and makes high-level strategic decisions (e.g., "Delay production batch A to trade flexibility on the intraday market") rather than just summarizing reports.*
Rationale for Joint R&D Project	***The Gap:** Current industrial systems are either purely data-driven (black box) or rule-based (rigid). There is a lack of systems that combine **Physical Precision** (German Deep Tech) with **Operational Agility** (Korean Manufacturing).* ***Synergy:** We provide the **Cognitive Software Stack** (PINNs, LLM-Agents, Optimization Algorithms). We need a Korean partner to develop a **Control Hardware (Edge Gateway)** and integrate the system into the **Live Control Loop** of a high-tech production line (e.g., Battery/Semiconductor) to validate the active intervention capabilities.*
Key Words	1. **Physics-Informed Neural Networks (PINNs)** 2. **Time Series Foundation Models & LLMs** 3. **Active Flexibility Control / Virtual Power Plant** 4. **Optimization Methods and Forecasting** 5. **Agentic AI**

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3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	**Start of the Project**	Level 4 (Component validation in laboratory environment)	**End of the Project**	Level 7 (System prototype demonstration in an operational environment)
R&D Objectives	• *PINN-based Diagnostics: Achieve >90% accuracy in localizing physical anomalies (e.g. leakages) by embedding physical constraints into the neural network loss functions.* • *Autonomous Flexibility Control: Develop a Deep Reinforcement Learning controller that optimizes energy costs and revenues from grid services autonomously against dynamic market signals.* • *Neuro-Symbolic Agent Interface: Create an LLM-based agent that can query Time Series Foundation Models to forecast scenarios and autonomously execute control decisions (Human-in-the-loop validation).*			
R&D Target Specifications	1. ***Control Latency:*** *Decision-to-Action time < 1 minute for frequency containment reserves (FCR).* 2. ***Optimization Gain:*** *Reduce energy costs by >20% through active load shifting and peak shaving compared to baseline.* 3. ***Model Generalization:*** *TSFMs capable of zero-shot forecasting on new sensor streams without extensive retraining.* 4. ***Reasoning Capability:*** *The LLM agent must provide "Chain-of-Thought" explanations for every control decision (Explainable AI).*			

R&D Plan	***Research Approach:*** *The project follows an agile* ***"Model-Based Design"*** *approach, evolving from offline algorithm training (Year 1) to Hardware-in-the-Loop integration (Year 2) and final deployment in a live industrial production environment (Year 3). We apply a* ***"Human-in-the-Loop"*** *validation strategy for the AI agents to ensure safety before enabling fully autonomous control.*
	Year 1: Foundation – Algorithmic Core & System Architecture
	• ***Task 1.1: Physics-Informed Modeling (PINNs):***
	  ○ *Development of base PINN models that integrate fluid dynamic laws (Navier-Stokes) into neural networks.*
	  ○ *Adaptation of models to specific Korean machinery types (e.g., compressors, chillers) provided by the partner.*
	• ***Task 1.2: Foundation Model Training:***
	  ○ *Pre-training of* ***Time Series Foundation Models (TSFMs)*** *on historical multi-modal datasets to enable zero-shot forecasting capabilities.*
	  ○ *Definition of the data schema for the* ***"Semantic Data Layer"*** *to allow the LLM to understand heterogeneous sensor data.*
	• ***Task 1.3: Hardware Specification (Joint Task):***
	  ○ *Definition of requirements for the* ***Industrial Edge Controller*** *(GPU specs, latency, interfaces) to support local inference of LLMs and PINNs.*
	• ***Milestone Y1:*** *Validated "Offline" Algorithms (PINNs & TSFM) & Finalized Hardware Architecture Specifications.*
	Year 2: Integration – The Cognitive Edge System
	• ***Task 2.1: Neuro-Symbolic Agent Construction:***
	  ○ *Development of the* ***LLM-Reasoning Engine*** *that orchestrates the TSFMs. Implementation of* ***RAG (Retrieval-Augmented Generation)*** *to link sensor anomalies with technical documentation.*
	  ○ *Construction of the AI agent for flexibility optimization, training it in a digital twin*			

simulation environment.

- ***Task 2.2: Edge-Porting & Optimization:***
  - *Quantization and pruning of the heavy AI models to run efficiently on the partner's Edge Hardware.*
- ***Task 2.3: Hardware Prototype Development (Korean Partner):***
  - *Manufacturing of the Edge Gateway prototype including interfaces to PLCs (OPC UA / Modbus).*
- ***Task 2.4: Lab-Scale Validation:***
  - ***Hardware-in-the-Loop (HiL)** testing: Running the AI algorithms on the physical prototype using simulated production signals to verify real-time capability (<1 min latency).*
- ***Milestone Y2:** Fully Integrated "Cognitive Energy Edge-Prototype" (Alpha Version) ready for field installation.*

Year 3: Deployment – Field Pilot & Autonomy

- ***Task 3.1: Installation in Operational Environment:***
  - *Deployment of the prototype in the Korean partner's testbed (e.g., semiconductor or battery supply chain facility).*
  - *Integration with live SCADA systems for read/write access.*
- ***Task 3.2: Phase 1 - Open-Loop Assistance:***
  - *System operates in "Advisor Mode": The LLM Agent suggests flexibility trades and leakage repairs; human operators validate decisions to build trust.*
- ***Task 3.3: Phase 2 - Closed-Loop Autonomy:***
  - *Activation of **Active Flexibility Control**: The system autonomously shifts loads and controls storage based on live market price signals (Virtual Power Plant operation).*
- ***Task 3.4: Impact Assessment & Commercialization:***
  - *Quantification of savings (Target: >20% cost reduction) and anomaly detection rates (>90%).*
  - *Development of a joint go-to-market strategy for the Asian and European markets.*
- ***Milestone Y3:** TRL 7 System Demonstration, Verified Economic Impact Report, and Commercial Rollout Plan.*

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4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	• *Development of Physics-Informed Neural Networks for component diagnosis.* • *Implementation of the "Cognitive Decision Layer": Coupling Time Series Foundation Models with LLM reasoning capabilities.* • *Design of AI agents for flexibility optimization.*
Expected R&R of Korean SME Partner	• *Development of the **Industrial Edge Controller** capable of running heavy AI models (LLM/TSFM) locally* • ***Shopfloor Integration:** Interfacing the AI Agent with legacy PLCs and SCADA systems for active control (Write-Access).* • *Provision of a dynamic production environment for testing flexibility scenarios (e.g., load shedding/shifting).*

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5. Expected Outcomes/Impact

Expected Outcomes and Impact	***Technological:*** *A paradigm shift from "Energy Monitoring" to* ***"Energy Autonomy"****. The system does not just show data, it understands physics and decides economically.* ***Economic:*** *Direct monetization of production flexibility. Korean SMEs can offer their machines as "Grid-Interactive" assets, opening new revenue streams beyond pure manufacturing.* ***Industrial:*** *Solving the "Skill Gap" in energy management. The LLM-Agent acts as an expert engineer, democratizing advanced energy optimization for factories without specialized personnel.*

6. References

(1) Core Competence: Anomaly Detection & Active Learning Holtz, D., **Kaymakci, C.**, Leuthe, D., Wenninger, S., & **Sauer, A.** (2025). A data-efficient active learning architecture for anomaly detection in industrial time series data. *Flexible Services and Manufacturing Journal*, 1-32.

(2) Core Competence: Deep Learning & Autoencoders in Industry Kaymakci, C., Wenninger, S., & **Sauer, A.** (2021). Energy anomaly detection in industrial applications with long short-term memory-based autoencoders. *Procedia CIRP*, *104*, 182-187.

(3) Core Competence: Compressed Air Diagnostics Wolf, C., Baur, L., Harman, A., **Kaymakci, C.**, & **Sauer, A.** (2024). Quantitative Characterization of Compressed Air Systems: A Methodological Approach to Analyzing Pressure Drop Curves. *Proceedings of the Conference on Production Systems and Logistics (CPSL)*.

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(5) Core Competence: Interpretation of AI Models Wenninger, S., **Kaymakci, C.**, & Wiethe, C. (2022). Explainable long-term building energy consumption prediction using QLattice. *Applied Energy*, *308*, 118300.

(6) Context: Industrial Flexibility & IT Integration Bauer, D., Hieronymus, A., **Kaymakci, C.**, Köberlein, J., Schimmelpfennig, J., & **Sauer, A.** (2021). Wie IT die Energieflexibilitätsvermarktung von Industrieunternehmen ermöglicht und die Energiewende unterstützt. *HMD Praxis der Wirtschaftsinformatik*, *58*(1), 102-115.

(7) Context: Federated Learning & Data Privacy Kaymakci, C., Baur, L., & **Sauer, A.** (2021). Federated machine learning architecture for energy-efficient industrial applications. *Hannover: publish-Ing.*

(8) Foundational Technology: Physics-Informed Neural Networks (PINNs) Raissi, M., Perdikaris, P., & Karniadakis, G. E. (2019). Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations. *Journal of Computational Physics*, *378*, 686-707.

☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of

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the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization